

20) Debug and Correct a CSS and JavaScript Web Application

Browser Developer Tools, console, Elements panel, Sources panel, breakpoints, console.log(), debugger, try...catch, CSS inspection and error correction.

Code

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Debugging Demo</title>
```

```
<style>
```

```
  * {
```

```
    box-sizing: border-box;
```

```
    margin: 0;
```

```
    padding: 0;
```

```
  }
```

```
  body {
```

```
    font-family: Arial, sans-serif;
```

```
    background: #eef2f7;
```

```
    min-height: 100vh;
```

```
    display: flex;
```

```
    justify-content: center;
```

```
    align-items: center;
```

```
  }
```

```
.container {  
  width: 90%;  
  max-width: 600px;  
  background: white;  
  padding: 30px;  
  border-radius: 15px;  
  box-shadow: 0 8px 25px rgba(0,0,0,0.2);  
  text-align: center;  
}
```

```
h1 {  
  color: #333;  
  margin-bottom: 20px;  
}
```

```
input {  
  width: 100%;  
  padding: 12px;  
  margin: 10px 0;  
  border: 2px solid #ccc;  
  border-radius: 8px;  
  font-size: 16px;  
}
```

```
button {  
  padding: 12px 25px;  
  margin: 10px 5px;
```

```
border: none;
border-radius: 8px;
background: #4f46e5;
color: white;
font-size: 16px;
cursor: pointer;
}
```

```
button:hover {
    background: #3730a3;
}
```

```
#output {
    margin-top: 20px;
padding: 15px;
background: #e8f5e9;
border-radius: 8px;
color: #2e7d32;
font-size: 18px;
font-weight: bold;
}
```

```
.error {
    background: #ffebee;
    color: #c62828;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h1>CSS & JavaScript Debugging App</h1>
```

```
<input type="number" id="num1" placeholder="Enter first number">
```

```
<input type="number" id="num2" placeholder="Enter second number">
```

```
<button id="addBtn">Add</button>
```

```
<button id="clearBtn">Clear</button>
```

```
<div id="output">Result will appear here</div>
```

```
</div>
```

```
<script>
```

```
const num1 = document.getElementById("num1");
```

```
const num2 = document.getElementById("num2");
```

```
const addBtn = document.getElementById("addBtn");
```

```
const clearBtn = document.getElementById("clearBtn");
```

```
const output = document.getElementById("output");
```

```
function calculateSum() {
```

```
  try {
```

```
    console.log("Calculation started");
```

```
debugger;

const value1 = Number(num1.value);
const value2 = Number(num2.value);

if (num1.value === "" || num2.value === "") {
    throw new Error("Please enter both numbers.");
}

const result = value1 + value2;

console.log("First number:", value1);
console.log("Second number:", value2);
console.log("Result:", result);

output.classList.remove("error");
output.innerHTML = "Result: " + result;
}

catch (error) {
    console.error("Error:", error.message);

    output.classList.add("error");
    output.innerHTML = error.message;
}

}

function clearFields() {
```

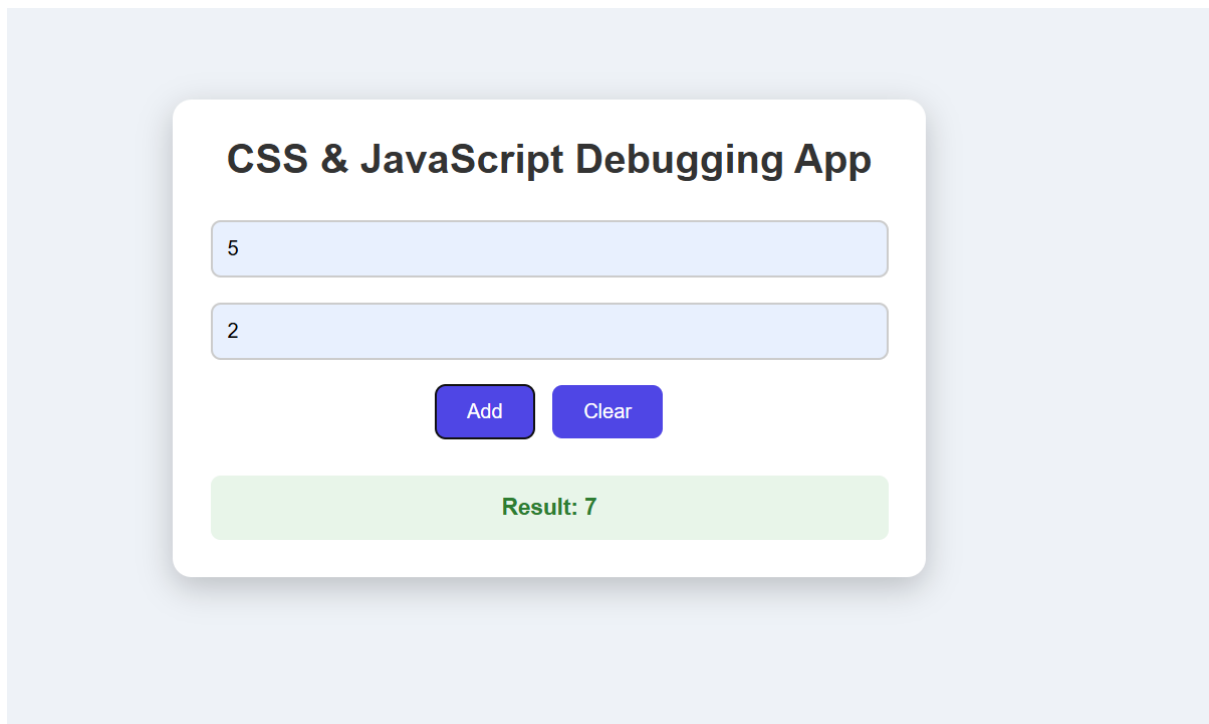
```
num1.value = "";
num2.value = "";
output.innerHTML = "Result will appear here";
output.classList.remove("error");

console.log("Fields cleared");
}

addBtn.addEventListener("click", calculateSum);
clearBtn.addEventListener("click", clearFields);
</script>

</body>
</html>
```

Output



The screenshot shows a web application interface with a light blue background. In the center is a white rounded rectangle with a subtle shadow. At the top of this rectangle is the title "CSS & JavaScript Debugging App" in bold black text. Below the title are two light blue input fields. The first field contains the number "5" and the second field contains the number "2". Underneath these fields are two blue buttons with white text: "Add" and "Clear". At the bottom of the white rectangle is a light green rounded rectangle containing the text "Result: 7" in bold green font.